Quick Reference - What Changed

🔄 Files to Update

1. src/blockchain.js - REPLACE ENTIRE FILE

- Wrapped in IIFE `(function() { ... })()`
- Removed ES6 dynamic imports
- Uses `window.CKBService` for all blockchain ops
- Creates real cells on CKB blockchain
- Proper lockscript sponsorship for voters

2. web/organizer.html - UPDATE SCRIPT TAGS ONLY

Replace the `<script>` section at the bottom with:

```html
<!-- Scripts: Load in correct order -->
<!-- 1. ES Module Shims -->
<script async src="https://ga.jspm.io/npm:es-module-shims@2/dist/es-module-shims.js"></script>

<!-- 2. Import Map -->
<script type="importmap">
{
  "imports": {
    "@joyid/ckb": "https://esm.sh/@joyid/ckb@1.1.2?bundle&target=es2022&polyfill-node",
    "@ckb-lumos/lumos": "https://esm.sh/@ckb-lumos/lumos@0.23.0?bundle&target=es2022&polyfill-node"
  }
}
</script>

<!-- 3. CKB Service Bridge (ES Module) -->
<script type="module" src="../src/ckbServiceBridge.js"></script>

<!-- 4. Blockchain Module (Regular script) -->
<script src="../src/blockchain.js"></script>

<!-- 5. Organizer Script (Regular script) -->
<script src="organizer.js"></script>
```

3. src/ckbServiceBridge.js - NO CHANGES NEEDED ✓

Already correct from previous version.

4. web/organizer.js - NO CHANGES NEEDED ✓

Already compatible with new blockchain.js.

📋 Quick Implementation Checklist

- [ ] Copy new `blockchain.js` content (IIFE-wrapped version)
- [ ] Update `organizer.html` script loading section
- [ ] Keep `ckbServiceBridge.js` as-is
- [ ] Keep `organizer.js` as-is
- [ ] Test in browser with F12 console open
- [ ] Verify both services load successfully
- [ ] Connect JoyID wallet
- [ ] Publish test election
- [ ] Check transaction on CKB explorer

🎯 Expected Console Output (Success)

```
✓ CKB Service Bridge loaded successfully
  Network: testnet
  RPC: https://testnet.ckb.dev/rpc
  Indexer: https://testnet.ckb.dev/indexer
✓ VoteSecure Blockchain Module loaded
  Network: testnet
  RPC: https://testnet.ckb.dev/rpc
VoteSecure Organizer loaded successfully
Debug interface available at: window.VoteSecureOrganizer
✓ CKB Service ready
```

🚀 Testing Steps

1. **Open organizer.html**
2. **Open browser console (F12)**
3. **Click "Connect JoyID"**
4. **Approve connection in JoyID popup**
5. **Create election (fill all steps)**
6. **Review & Publish**

7. **Approve transaction in JoyID popup**
8. **Check success modal for txHash**
9. **Click explorer link to verify on blockchain**

💡 Key Differences from Old Version

Old blockchain.js	New blockchain.js
Returns `success: true` but doesn't create cells	Actually creates cells on CKB blockchain
No JoyID integration	Uses JoyID for signing transactions
Simulated transactions	Real blockchain transactions
No gas sponsorship	Lockscript sponsors voter transactions
No balance checking	Validates sufficient CKB before tx
No transaction hashes	Returns real txHash + explorer URLs

🔗 Important Links

- **Testnet Explorer**: https://pudge.explorer.nervos.org/
- **Testnet Faucet**: https://faucet.nervos.org/
- **JoyID**: https://testnet.joyid.dev/

One-Line Summary

Before: Simulated blockchain with local storage
After: Real CKB blockchain transactions with JoyID wallet integration